

Periodic Scalar Play as a Sweeping Process: Exact Clamp, Entrainment, and Dissipation

Route-A source-local certificate HCS-C332

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Abstract

For a scalar Moreau play element driven by one periodic rise and fall, we prove the *segment projection* and *exact Poincaré clamp* for every $W^{1,1}$ input, including plateaus and corners. The *three chambers* and *one-period entrainment* follow from one idempotent map, together with order, nonexpansion, and rate independence.

1 Segment solution and return map

Fix $r \geq 0$. Let $u \in W^{1,1}([0, T])$ satisfy $u(0) = u(T) = m$. For some τ , assume that u is nondecreasing from m to M on $[0, \tau]$ and nonincreasing back to m on $[\tau, T]$. Plateaus are allowed. Put $D = M - m$ and

$$C(t) = [u(t) - r, u(t) + r].$$

We use $N_C(y) = \{v : v(z - y) \leq 0 \text{ for all } z \in C\}$ and solve

$$y(t) \in C(t), \quad -\dot{y}(t) \in N_{C(t)}(y(t)) \quad \text{a.e.} \quad (1)$$

Theorem 1 (Projection and Poincaré clamp). *For every $z \in [m - r, m + r]$, problem (1) with $y(0) = z$ has a unique absolutely continuous solution. On a monotone segment beginning at t_0 ,*

$$y(t) = \text{Proj}_{[u(t)-r, u(t)+r]} y(t_0). \quad (2)$$

The return map from one minimum to the next is

$$P(z) = \min\{m + r, \max\{M - r, z\}\}. \quad (3)$$

Proof. On a nondecreasing segment define $y(t) = \max\{y(t_0), u(t) - r\}$. It is feasible. Before the lower endpoint reaches the state, $\dot{y} = 0$; when the endpoint pushes it, $y = u - r$, $\dot{y} = \dot{u} \geq 0$, and $-\dot{y}$ lies in the lower-endpoint normal cone. Feasibility and the normal-cone sign force these alternatives for any solution, proving uniqueness. On a nonincreasing segment the corresponding formula is $y(t) = \min\{y(t_0), u(t) + r\}$. Both formulas are (2) and remain valid almost everywhere across plateaus and corners.

The rise ends at $z_\uparrow = \max\{z, M - r\}$. The fall ends at $\min\{z_\uparrow, m + r\}$, proving (3). $\square$

2 Three chambers and finite-time entrainment

Theorem 2 (Complete periodic atlas). *The map P is increasing, one-Lipschitz, and idempotent. Every admissible initial state therefore reaches a periodic response in at most one period. Its fixed responses are as follows.*

1. *If $D < 2r$, the fixed set is $[M - r, m + r]$, and each fixed response is constant.*
2. *If $D = 2r$, this interval collapses to the single constant response $m + r = M - r$.*
3. *If $D > 2r$, the unique fixed state is $m + r$ and it generates the unique nonconstant periodic loop.*

For a common input, the solution maps preserve initial order and satisfy $|y_1(t) - y_2(t)| \leq |y_1(0) - y_2(0)|$.

Proof. When $D < 2r$, equation (3) is scalar projection onto $[M - r, m + r]$; at equality the interval is a singleton. When $D > 2r$, $M - r > m + r$, so (3) is the constant $m + r$. This proves all fixed sets and directly gives $P^2 = P$. A fixed point in the first two chambers is never pushed through a positive distance. In the third chamber, it moves from $m + r$ to $M - r$ and back. Projection on an interval is increasing and one-Lipschitz; composing the two segment maps proves the last assertion. $\square$

If ϕ is an orientation-preserving absolutely continuous surjection and both u and $u \circ \phi$ lie in $W^{1,1}$, formula (2) depends only on the ordered input value. Uniqueness therefore gives

$$\mathcal{P}_r[u \circ \phi] = \mathcal{P}_r[u] \circ \phi.$$

This is the precise admissible reparameterization statement; no inadmissible composition is silently included.

Source lineage

This source-local reconstruction makes no priority claim.

References

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- [2] M. A. Krasnosel’skii and A. V. Pokrovskii, *Systems with Hysteresis*, Springer, 1989. DOI: [10.1007/978-3-642-61302-9](https://doi.org/10.1007/978-3-642-61302-9).
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